

On Using Fiber Transfer Functions to Characterize Laser Chirp and Fiber Dispersion

R. C. Srinivasan and J. C. Cartledge, *Member, IEEE*

Abstract—Two measurement techniques for estimating the α parameter of a laser, the frequency f_c for which the adiabatic and transient chirp of a laser have the same magnitude, and the dispersion coefficient D of an optical fiber are compared. The techniques rely on either a single measurement of the transfer function of a dispersive optical fiber using direct modulation of the laser, or two measurements of the fiber transfer function using external and direct modulation of the laser. Compared to the two measurement technique, the single measurement technique yields a smaller estimate of α and a nearly identical estimate of f_c , and can produce an unreliable estimate of D unless the frequency span in the parameter extraction procedure is sufficiently large.

I. INTRODUCTION

NETWORK analysis based measurement techniques for estimating the transient and adiabatic chirp of a laser and the dispersion coefficient of an optical fiber have recently been presented. The dispersion coefficient D can be accurately determined from measurement of the transfer function of a dispersive fiber using a CW laser and lithium niobate Mach-Zehnder modulator with low adiabatic chirp [1], [2]. The parameter extraction procedure uses sharply defined resonance frequencies in the measured transfer function. When the transfer function of a dispersive fiber is measured using direct modulation of a laser with both adiabatic and transient chirp, the dispersion coefficient, α parameter (or linewidth enhancement factor) of the laser, and frequency f_c for which the adiabatic and transient chirp of the laser have the same magnitude can be determined [3]. The parameter extraction procedure uses a three parameter curve fit of the calculated transfer function to the measured transfer function. In this paper, we compare two parameter extraction schemes. The first scheme, as described in [3], uses a three parameter curve fit (α , D and f_c) to the fiber transfer function measured using direct modulation of the laser. The second scheme, described here for the first time, uses the approach in [1] to determine D from the transfer function measured using external modulation of the laser, and a two parameter curve fit (α and f_c) to the transfer function measured using direct modulation of the laser. In particular, we investigate the dependence of the parameter value estimates on the frequency span over which the curve fitting is performed.

Manuscript received June 8, 1995; revised July 14, 1995. This work was supported in part by Bell-Northern Research, the Natural Sciences and Engineering Research Council of Canada, and the Telecommunications Research Institute of Ontario.

The authors are with the Department of Electrical and Computer Engineering, Queen's University, Kingston, ON K7L 3N6 Canada.

IEEE Log Number 9414886.

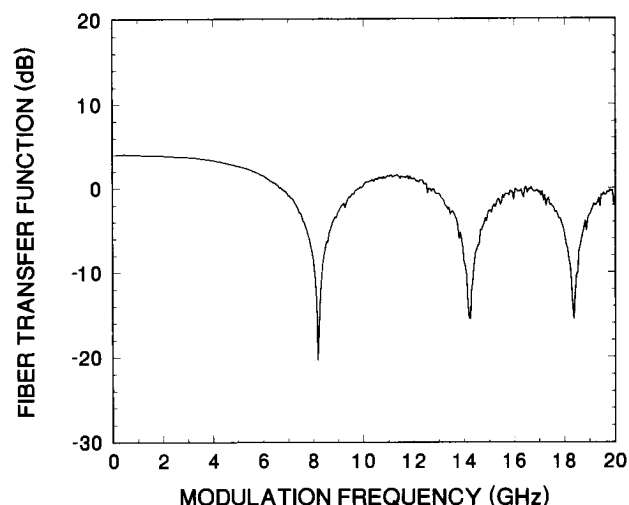


Fig. 1. Measured transfer function of a dispersive optical fiber using an externally modulated laser.

II. RESULTS AND DISCUSSION

The dispersion coefficient D is determined from the transfer function of 57.2 km of standard single-mode optical fiber, measured using a lightwave component analyzer (HP 8703) with a CW optical signal from the laser under test applied externally to the input of the Mach-Zehnder modulator. The influence of the transmitter and receiver in the analyzer is cancelled by dividing the response with the optical fiber by the response without the optical fiber. An erbium doped fiber amplifier (EDFA) is used to offset the fiber loss. The transfer function is shown in Fig. 1 for a multiple-quantum-well (MQW) DFB laser mounted in a butterfly package for 2.5 Gb/s operation. The laser threshold and bias currents are 18 mA and 45 mA, respectively. The dispersion coefficient, which is determined from the sharply defined resonance frequencies, is 16.13 ps/(km·nm) at a wavelength of 1554.42 nm with a standard deviation of 0.07 ps/(km·nm).

The fiber transfer function is also measured using direct modulation of the MQW-DFB laser. In this case both the adiabatic and transient chirp play important roles. The measurement technique is identical to that described above with external modulation of the laser replaced by direct modulation of the laser. In this case, the normalized small signal transfer function is given by [3]–[5]

$$H(f) = \cos(\theta) - (\alpha - j\alpha f_c/f) \sin(\theta) \quad (1)$$

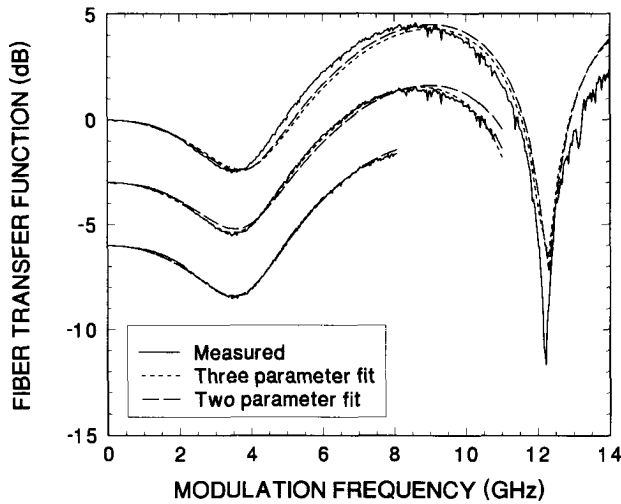


Fig. 2. Measured and calculated transfer functions of a dispersive optical fiber using a directly modulated laser. The calculated results are for a three-parameter fit of $H(f)$, and a two-parameter fit of $H(f)$ with D determined from Fig. 1. The frequency spans used in the curve fitting are 0.13 GHz to 8 GHz, 0.13 GHz to 11 GHz, and 0.13 GHz to 14 GHz. For clarity, the results for 0.13 GHz to 8 GHz are offset by -6 dB and the results for 0.13 GHz to 11 GHz are offset by -3 dB.

where

$$\theta = f^2 \frac{\pi \lambda^2 D L}{c} \quad (2)$$

and α , f_c , and D are as defined above, c is the speed of light, λ is the wavelength of the CW laser, and L is the fiber length. For a given fiber length and laser wavelength, the fitting of (1) to measured results either uses three parameters, α , f_c , and D , or two parameters, α and f_c , with D having been determined using the transfer function measured with the externally modulated laser.

Fig. 2 illustrates measured and calculated results for the fiber transfer function using the directly modulated MQW-DFB laser. The laser bias current is 45 mA and the modulation current is 1.125 mA. Due to the adiabatic chirp of the laser, the transfer function with direct modulation (Fig. 2) differs from that obtained with external modulation (Fig. 1). The measurement technique is very effective in cancelling the bandwidth limitation of the laser package. The frequency spans over which the curve fitting is performed are 0.13 GHz to 8 GHz, 0.13 GHz to 11 GHz, and 0.13 GHz to 14 GHz. The measured results are based on one swept frequency measurement from 0.13 GHz to 14 GHz. As expected, the agreement between the calculated and measured results for the two parameter curve fit is not quite as good as the agreement for the three parameter curve fit. The results obtained from the minimum mean square error curve fitting are summarized in Table I using different frequency spans for the fitting. For the three parameter curve fit, the values of α and f_c are reasonably consistent in contrast to the values of D , which vary considerably, but approach the estimate obtained from Fig. 1 as the frequency span increases. The extent of the dips in $10 \log |H(f)|$ is determined primarily by f_c and to a lesser extent by α and D . The frequency of the first dip is determined primarily by α , and to a lesser extent by

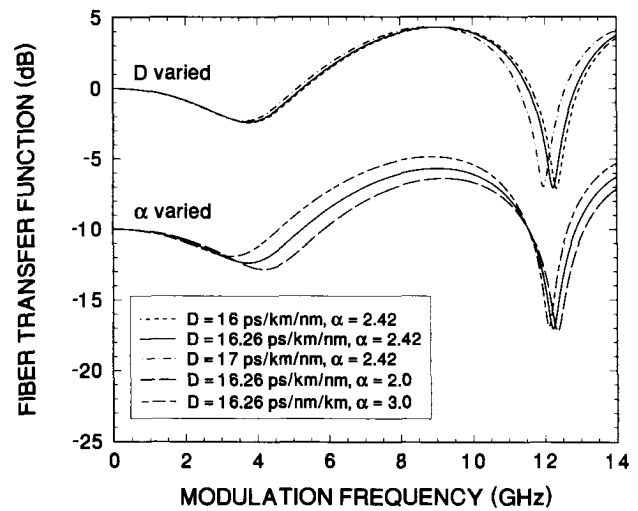


Fig. 3. Calculated transfer functions of a dispersive optical fiber using a directly modulated laser. $\alpha = 2.42$ for the cases where D is varied; $D = 16.26$ ps/(km·nm) for the case where α is varied. In both cases $f_c = 2.64$ GHz (see Table I). The results for α varied are offset by -10 dB for clarity.

TABLE I
RESULTS OF THE PARAMETER EXTRACTION FOR A LASER BIAS CURRENT OF 45 mA

Range (GHz)	Three parameter fit			Two parameter fit	
	α	f_c (GHz)	D (ps/km/nm)	α	f_c (GHz)
0.13-6.0	2.38	2.44	18.93	2.82	2.41
0.13-7.0	2.65	2.42	17.09	2.82	2.41
0.13-8.0	2.53	2.42	17.87	2.80	2.44
0.13-9.0	2.49	2.42	18.15	2.75	2.49
0.13-10.0	2.54	2.43	17.76	2.70	2.56
0.13-11.0	2.59	2.47	17.19	2.64	2.65
0.13-12.0	2.58	2.63	16.56	2.67	2.59
0.13-13.0	2.54	2.46	16.24	2.68	2.35
0.13-14.0	2.42	2.64	16.26	2.56	2.53
Average	2.52	2.48	17.34	2.72	2.49
Standard deviation	0.08	0.08	0.86	0.08	0.09

D , whereas the frequency of the second dip is determined primarily by D and to a lesser extent by α . This is illustrated in Fig. 3 by separately varying α and D about the extracted values obtained for the 0.13 GHz to 14 GHz frequency span. Consequently, it is necessary for the upper frequency to exceed the frequency of the second dip (12.3 GHz) in order to obtain a reasonable estimate of D . For the two parameter curve fit with $D = 16.13$ ps/(km·nm), the values of α and f_c are both reasonably consistent.

Results were also obtained for laser bias currents of 35 mA and 25 mA (Tables II and III, respectively). In all cases, the three parameter curve fitting yields results for D that depend on the frequency range over which the fitting is performed. To provide further confirmation, measurements were performed on a MQW-DFB chip-on-carrier laser which is wire bonded to a coplanar waveguide substrate. This configuration has a mount-limited bandwidth of 10.5 GHz. Results for a bias current of 45 mA and a modulation current of 1.125 mA are summarized in Table IV and are in agreement with those presented in Tables I-III. In each of the four cases, the estimates of α show less variation for larger frequency spans and the estimates of f_c show less variation for smaller frequency spans. Also, better agreement between the values of D obtained by external and direct (0.13 GHz to 14 GHz frequency span) modulation is accompanied by better agreement between the

TABLE II
RESULTS OF THE PARAMETER EXTRACTION FOR A LASER BIAS CURRENT OF 35 mA

Range (GHz)	Three parameter fit			Two parameter fit	
	α	f_c (GHz)	D (ps/km/nm)	α	f_c (GHz)
0.13-6.0	1.77	1.31	24.72	2.85	1.23
0.13-7.0	1.68	1.32	25.84	2.81	1.26
0.13-8.0	1.94	1.30	22.73	2.76	1.29
0.13-9.0	2.12	1.29	20.85	2.70	1.34
0.13-10.0	2.28	1.30	19.28	2.64	1.39
0.13-11.0	2.41	1.33	18.02	2.57	1.46
0.13-12.0	2.45	1.51	16.78	2.60	1.43
0.13-13.0	2.42	1.31	16.36	2.69	1.18
0.13-14.0	2.29	1.45	16.42	2.55	1.34
Average	2.15	1.35	20.11	2.69	1.32
Standard deviation	0.27	0.07	3.42	0.10	0.09

TABLE III
RESULTS OF THE PARAMETER EXTRACTION FOR A LASER BIAS CURRENT OF 25 mA

Range (GHz)	Three parameter fit			Two parameter fit	
	α	f_c (MHz)	D (ps/km/nm)	α	f_c (MHz)
0.13-6.0	2.20	435.0	21.20	2.98	422.0
0.13-7.0	2.43	429.9	19.40	2.98	422.9
0.13-8.0	2.63	426.8	18.07	2.97	424.0
0.13-9.0	2.72	425.8	17.48	2.97	425.5
0.13-10.0	2.80	425.4	17.01	2.96	427.4
0.13-11.0	2.87	426.0	16.60	2.96	429.2
0.13-12.0	2.93	429.7	16.18	2.96	428.7
0.13-13.0	2.93	412.1	16.08	2.95	408.7
0.13-14.0	2.91	418.5	16.08	2.94	414.4
Average	2.71	425.5	17.57	2.96	423.3
Standard deviation	0.24	6.3	1.65	0.01	6.9

values of α obtained by the two and three parameter curve fitting procedures.

III. SUMMARY

Two measurement techniques for estimating α , f_c , and D have been compared. The three parameter curve fitting procedure based on a single measurement of the fiber transfer function using direct modulation of the laser yields a good fit and consistent estimates of α and f_c , but only yields a reasonable estimate of D if the frequency span for the curve fitting includes the first and second dip in the magnitude of the transfer function. By comparison, the two parameter curve fitting procedure based on a measurement of the fiber transfer function using direct modulation of the laser to determine α and f_c , with D having been accurately determined by a separate measurement of the transfer function using external modulation of the laser, yields a slightly poorer fit, a larger consistent estimate of α and a nearly identical consistent esti-

TABLE IV
RESULTS OF THE PARAMETER EXTRACTION FOR A SECOND LASER WITH A BIAS CURRENT OF 45 mA

Range (GHz)	Three parameter fit			Two parameter fit	
	α	f_c (GHz)	D (ps/km/nm)	α	f_c (GHz)
0.13-6.0	2.23	2.71	24.28	3.37	2.68
0.13-7.0	2.46	2.71	22.01	3.29	2.75
0.13-8.0	2.74	2.72	19.61	3.23	2.81
0.13-9.0	2.81	2.73	19.02	3.14	2.92
0.13-10.0	2.89	2.77	18.23	3.03	3.07
0.13-11.0	2.94	2.89	17.30	2.93	3.22
0.13-12.0	2.93	3.03	16.63	3.06	2.98
0.13-13.0	2.85	2.93	16.32	3.07	2.76
0.13-14.0	2.76	3.05	16.34	2.97	2.90
Average	2.73	2.84	18.86	3.12	2.90
Standard deviation	0.23	0.13	2.59	0.14	0.16

mate of f_c . While the technique presented in [3] using direct modulation of the laser introduces a very useful parameter extraction procedure, the frequency span must be extended to include the second dip in the magnitude response if the extraction of unreliable estimates of D is to be avoided. Alternatively, the use of both external and direct modulation of the laser yields reliable estimates and a useful comparison.

ACKNOWLEDGMENT

The authors are grateful to H. Willemson of Bell-Northern Research for providing the packaged DFB laser, B. Villeneuve of Bell-Northern Research and A. P. Freundorfer of Queen's University for assistance in mounting the chip-on-carrier laser, Lycom A/S for providing the erbium-doped fiber for the EDFA, and the reviewers for their useful comments.

REFERENCES

- [1] B. Christensen, J. Mark, G. Jacobsen, and E. Bødtker, "Simple dispersion measurement technique with high resolution," *Electron. Lett.*, vol. 29, pp. 132-134, 1993.
- [2] F. Devaux, Y. Sorel, and J. F. Kerdiles, "Simple measurement of fiber dispersion and of chirp parameter of intensity modulated light emitter," *J. Lightwave Technol.*, vol. 11, pp. 1937-1940, 1993.
- [3] A. Røyset, L. Bjerkan, D. Myhre, and L. Hafskjær, "Use of dispersive optical fibre for characterisation of chirp in semiconductor lasers," *Electron. Lett.*, vol. 30, pp. 710-712, 1994.
- [4] B. Wedding, "Analysis of fibre transfer function and determination of receiver frequency response for dispersion supported transmission," *Electron. Lett.*, vol. 30, pp. 58-59, 1994.
- [5] J. Wang and K. Petermann, "Small signal analysis for dispersive optical fiber communication systems," *J. Lightwave Technol.*, vol. 10, pp. 96-100, 1992.